AMITY UNIVERSITY
MADHYA PRADESH
Established vide Government of Madhya Pradesh Act No. 27 of 2010

Syllabus

Programme Name: B. Tech. (CSE/ECE/IT)		**Session:** 2023-27
Course Code: CSE603	**Course Name:** Internet of Things	**Semester:** VI

Credits (Total)	L	T	P	Marks (Internal/External		Contact Hours (per week)	Independent Study Hour (per week)	Section (Group)
3	2	1	0	30	70	3	3	
UG level						**Basic and applied**	**Student specific course outcome**	**Higher Education Placement Research**

Course Objective:

The objective of the course is to Introduction of IOT, Understand IOT Market perspective, Data and Knowledge Management and use of Devices in IOT Technology, Understand State of the Art – IOT Architecture. Real World IOT Design Constraints, Industrial Automation and Commercial Building Automation in IOT.

Course outcomes: After completion of course, the student will be able to:

CO-1	Understand the fundamental concept of Python in Eclipse Study, Install and Demonstrate basic operations
CO-2	Implement the different types of Arduino, Install IDE and perform basic LED programs.
CO-3	Demonstrate RFID, NFC and MQTT.
CO-4	Analyze the differences between Arduino and Raspberry Pi and Demonstrate Raspberry Pi basic LED programs and Zigbee Protocol.
CO-5	Design basic solutions using Arduino/Rasberry Pi and sensors

Teaching Pedagogy:

T1	Classroom teaching (white board), Power Point Presentations, Interactive lectures, Inquiry based teaching
T2	ABL activities, Assignments, Flip Class/ Seminars, Quiz, Oral Viva-voce examination

Assessment Tools

AT1-1	Quiz
AT1-2	Activity Based Learning
AT1-3	Midterm Exams
AT1-4	Flip Class
AT1-5	Seminar Presentation
AT1-6	Assignments
AT1-7	Poster
AT1-8	Oral Viva-voce examination
AT1-9	Industrial Visit Report

Prerequisites: HSC Standard Mathematics, knowledge of algebraic operations (manipulating equations, solving linear equations etc)

Module wise contents details	Assessment tools
Module I: Introduction to the Internet of Things: (9 Hours) Key Features, advantages, disadvantages, Wearable electronics, The Basics of Sensors & Actuators, Introduction to Cloud Computing, IOT Software.	Quiz Mid-term Exam Assignment
Module II: IoT-An Architectural Overview: (9 Hours) Building an architecture, Main design principles and needed capabilities, An IoT architecture outline, standards considerations. IoT Technology Fundamentals-Devices and gateways, Local and wide area networking, Data management, Business processes in IoT, Everything as a Service (XaaS), M2M and IoT Analytics, Knowledge Management.	Mid-Term Quiz Assignment
Module III: The Arduino Platform: (9 Hours) What is Arduino, Why Arduino, Driver installation, programming & Burning, Coding in wiring language, Compiling in Arduino, The Arduino Open-Microcontroller Platform, Arduino Basics, Arduino Board Layout & Architecture Reading from Sensors.	Mid-Term Oral Viva-voce examination Seminar Presentation
Module IV: Arduino Programming & Interface of Sensors: (9 Hours) LED display, PUSH button to array of LED, Communicating to and from computer, GSM, GPS and Zigbee interfacing, Interface sensor with arduino, Programming arduino, Reading from sensor, Connecting Arduino with Mobile Device. The Android Mobile OS, Using the Bluetooth Module.	Quiz Assignment Industrial Visit Report Seminar Presentation
Module V: Projects: (9 Hours) 1. Creating own Android App using MIT App Inventor & controlling Arduino connected devices. 2. Use Arduino to Upload free data from Environmental Sensors to Cloud Server. 3. Receive Automatic Call Notification on Mobile Phone for Burglar Alarm using IoT Platform 4. Control Electronic Devices from anywhere across the world using Internet & Mobile App.	Quiz Assignment Industrial Visit Report Poster Oral Viva-voce examination

Additional Learning:

List of Assignments	ASSIGNMENT -1 Q.1) Explain internet of things with key features, advantage and disadvantage with IOT software's. Q.2) Explain different types of IOT enabled wearable devices with specification. Q.3) Explain five smart IOT sensors and actuators with their characteristics, working principle, applications, advantage and disadvantages with their diagrams. ASSIGNMENT -2 Q.1) Describe in detail about IOT architecture with given below points: I. functional block diagram II. logical block diagram III. IOT levels IV. Communication models. Q.2) Explain the given term in respect of internet of things: I. data management II. business processing III. knowledge management ASSIGNMENT -3 Q.1) (a) Determine the importance of each layer in the generic M2M System Solution with a neat diagram. (b) Compare the pros and cons and identify different challenges of XaaS and IOT. Explain various trend in Information and communication technologies and its impact on IOT. Q.2) Illustrate the different component of Burglar Alarm. Describe the use of Burglar Alarm in various IOT applications. Calculate the cost of such system. ASSIGNMENT -4 Q.8) (a) Describe the Real-life IOT: five examples of industries and their importance of the Internet of Things for Increased Autonomy and Agility in Collaborative Production Environments? (b) Discuss the benefits and challenges of IOT during pandemic situation (COVID-19) in India. Explain how can be achieve the solution of these challenges. Q.9) Describe the various components of Arduino board and types of Arduino board with suitable diagrams. ASSIGNMENT -5 Q.1) Write short notes: I. cloud computing II. XaaS III. Iot and M2M IV. Types of sensors

	ASSIGNMENT -6 Q.1.) Explain the importance of wearable devices in healthcare system. Illustrate the types of wearable devices? Q2. Explain the different component of Burglar Alarm. Describe the use of Burglar Alarm in various IoT applications. Q3. Explain detail design guidelines of internet of thing. ASSIGNMENT -7 Q.1) Explain how to Connect Arduino with Mobile Device. The Android Mobile OS, Using the Bluetooth Module. Q.2 Creating own Android App using MIT App Inventor & controlling Arduino connected devices. ASSIGNMENT -8 Q1. Study and Configure Raspberry Pi. Q2. Explain and write the program for LED blink using Raspberry Pi. Q3. Study the Temperature sensor and Write Program for monitor temperature using Arduino ASSIGNMENT -9 Q.1) Write short notes: I. Comparisons in different Arduino Board II. Compare different temperature sensors Q2. Study and Implement Zigbee Protocol using Arduino / Raspberry Pi ASSIGNMENT -10 Q1. Explain the use Arduino to Upload free data from Environmental Sensors to Cloud Server. Q2. Explain working of Receive Automatic Call Notification on Mobile Phone for Burglar Alarm using IoT Platform Q3. Implement the Control Electronic Devices from anywhere across the world using Internet & Mobile App.
Suggested reading:	**Text:** • Jan Holler, VlasiosTsiatsis, Catherine Mulligan, Stefan Avesand, StamatisKarnouskos, David Boyle, **"From Machine-to-Machine to the Internet of Things: Introduction to a New Age of Intelligence",** 1st Edition, Academic Press, 2014. **Reference Books:** • Vijay Madisetti and ArshdeepBahga, **"Internet of Things (A Hands-on-Approach)",** 1stEdition, VPT, 2014. • Francis daCosta, **"Rethinking the Internet of Things: A Scalable Approach to Connecting Everything",** 1st Edition, Apress Publications, 2013
Suggested e- resources (Websites/e-books)	https://www.arduino.cc/en/software https://www.raspberrypi.org/

Assessment Plan:

Component of Evaluation	**Description**	**Code**	**Weightage %**
Continuous Internal Evaluation	Mid Term	CT	15%
	Seminar/Viva-Voce/Quiz/Home Assignment	S/V/Q/HA	10%
Attendance	A minimum of 75% Attendance is required to be maintained by a student to be qualified for taking the End Semester examination. The dispensation of 25% includes all types of leaves. including medical leaves.	A	5%
End Semester Examination	End Semester Examination	ESE	70%
Total			**100%**

Abbreviations: CT: Class Test, HA: Home Assignment, S/V/Q: Seminar/Viva/Quiz, ESE: End Semester Examination; A: Attendance

Course Articulation Matrix (Mapping of COs with POs)

CO	**CORRELATION WITH PROGRAMME OUTCOMES**												**CORRELATION WITH PROGRAMME SPECIFIC OUTCOMES**		
	PO 1	**PO 2**	**PO 3**	**PO 4**	**PO 5**	**PO 6**	**PO 7**	**PO 8**	**PO 9**	**PO 10**	**PO 11**	**PO 12**	**PSO1**	**PSO2**	**PSO3**
CO1	3	2	1						1						
CO2	3	2	1						1				1		
CO3	2								1						
CO4	3	2							1						
CO5	2		1						1			1	1		

1: strongly related, 2: moderately related and 3: weakly related